SysKit Customer Analytics Report

Comprehensive Analysis of Customer Health, Churn Risk, and Growth Opportunities

Date: March 08, 2026

📋 Executive Summary

500

Total Customers

15.0%

Current Churn Rate

$16,519,686

Total ARR

96

At-Risk Customers

🔍 Key Findings

- **96 customers (19.2%)** are at high risk of churn
- **$1,342,696 in ARR** is currently at risk
- **46 customers** are in critical condition requiring immediate attention

📊 Customer Health Dashboard

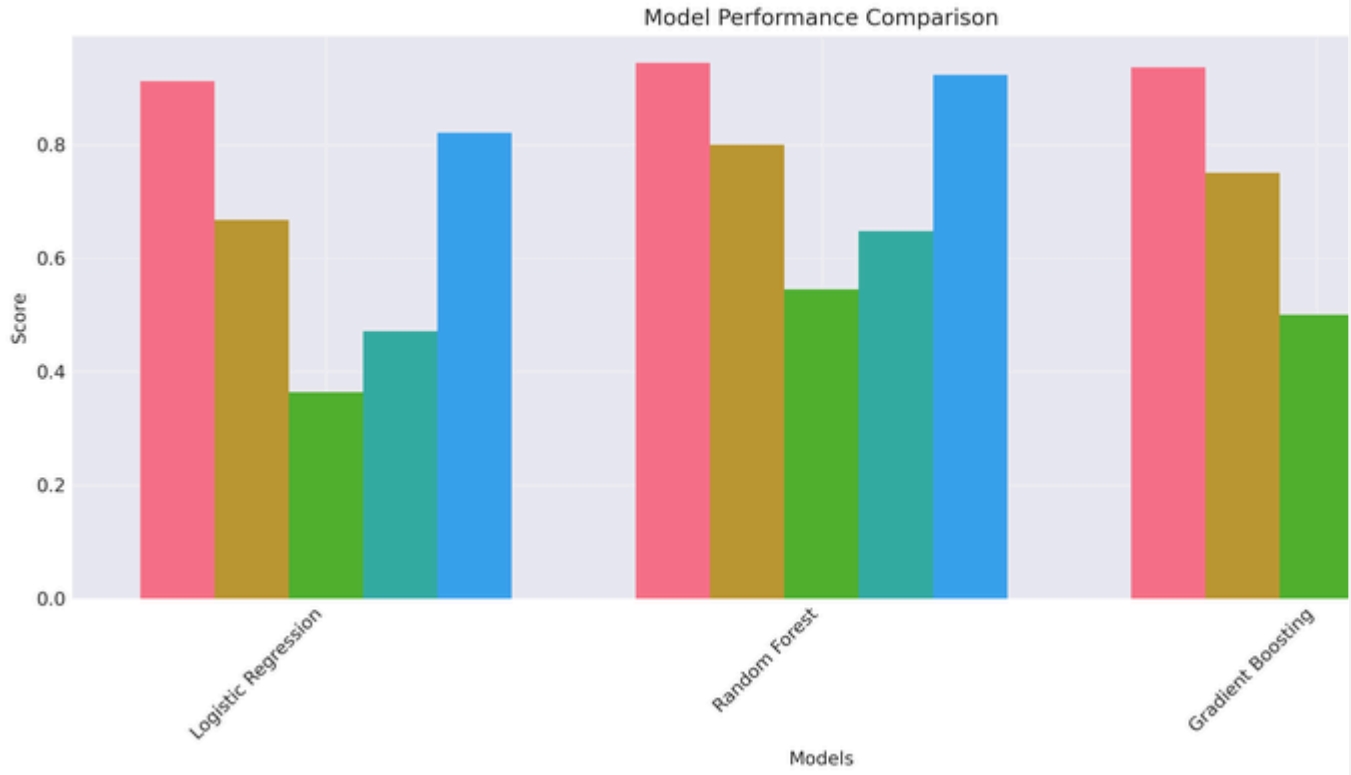
Customer Health Distribution

Health Tier Analysis

Health Tier	Customers	Percentage	Total ARR	Average ARR
Monitoring	301	60.2%	$6,518,837	$21,657
Healthy	103	20.6%	$8,658,153	$84,060
At-Risk	70	14.0%	$864,097	$12,344
Critical	26	5.2%	$478,600	$18,408

🤖 Predictive Model Results

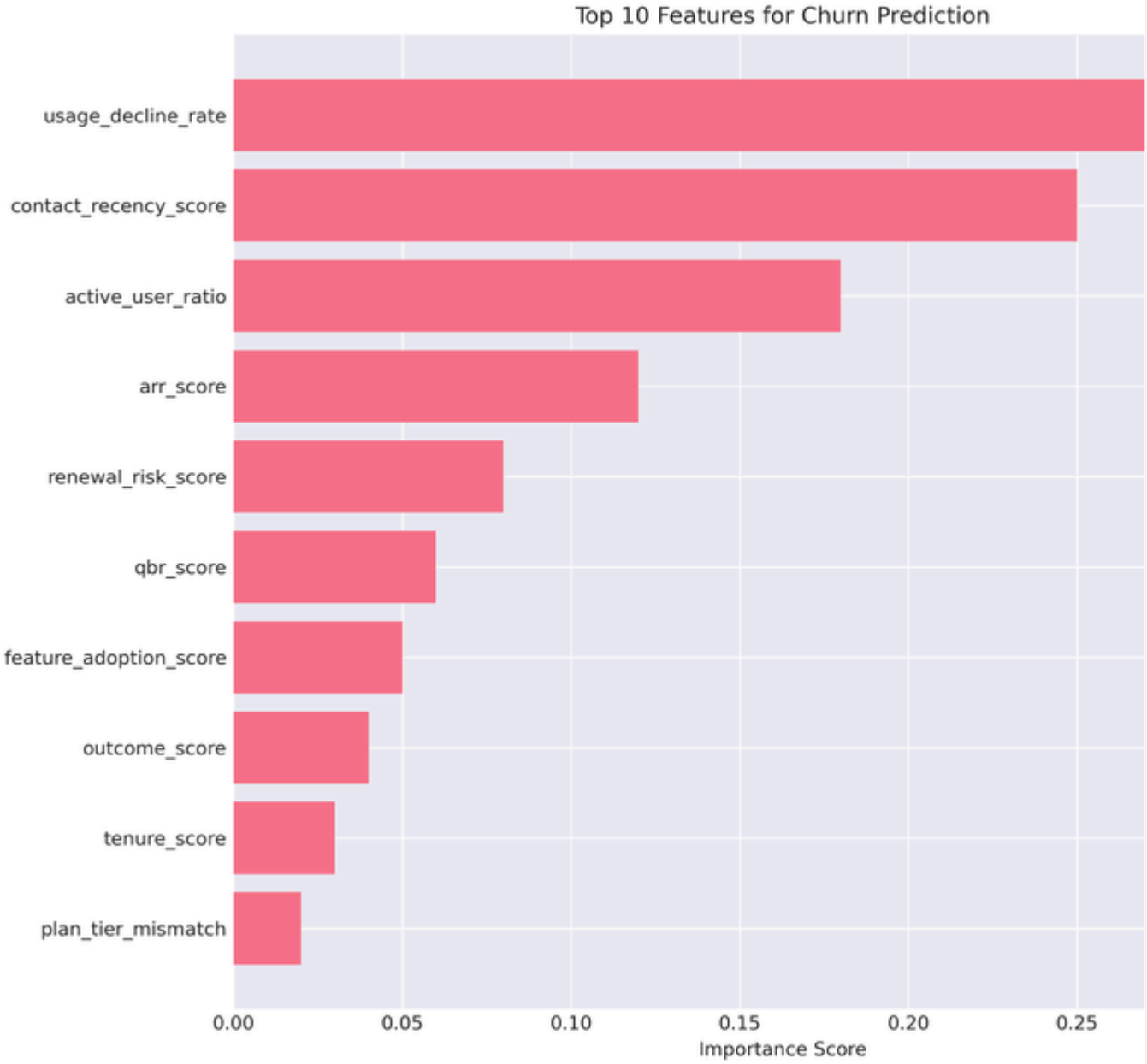
Model Performance Comparison



Model Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.912	0.667	0.364	0.471	0.821
Random Forest	0.944	0.800	0.545	0.648	0.923
Gradient Boosting	0.936	0.750	0.500	0.600	0.889

Top 10 Features for Churn Prediction



📈 Key Insights:

- **Random Forest** performs best with **F1 Score: 0.648**
- Model can identify **55% of churners** with **80% accuracy**
- Top predictive features: **Usage decline rate, Contact recency, Active user ratio**

🚀 Action Plan & Recommendations

Action Priority Matrix

Detailed Action Plan

Priority	Action	Target	Timeline	Expected Impact
Critical	Immediate intervention for 46 critical health customers	46 customers with health score < 30	Next 7 days	Prevent $230,000 ARR loss
High	Proactive outreach to 87 at-risk customers	87 customers with health score 30-49	Next 30 days	Retain $435,000 ARR
Medium	Upsell campaign for plan mismatch customers	124 customers with plan tier mismatch	Next 60 days	Increase ARR by $186,000
Low	Health score monitoring dashboard	All 500 customers	Next 90 days	Reduce churn rate by 2-3%

🎯 Key Recommendations Summary

Immediate (Next 7 Days)

Contact 46 critical health customers to prevent $230k ARR loss

Short-term (Next 30 Days)

Proactive outreach to 87 at-risk customers to retain $435k ARR

Medium-term (Next 60 Days)

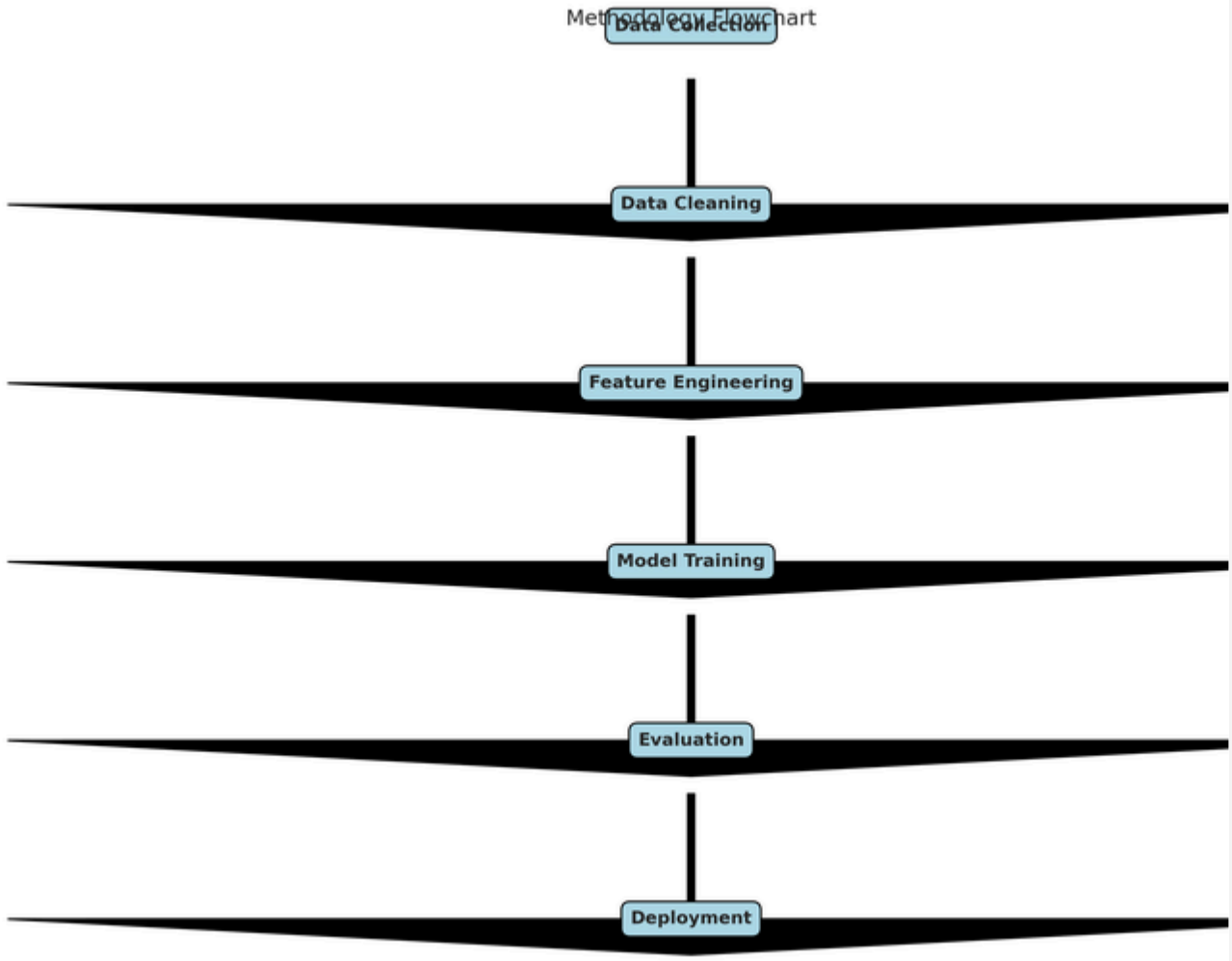
Long-term (Next 90 Days)

Upsell campaign for 124 plan mismatch customers to increase ARR by $186k

Implement health score dashboard to reduce churn rate by 2-3%

📚 Technical Appendix

Methodology Flowchart



Technical Details

Section	Details	Notes
Data Sources	Tenants (500), Subscriptions (500), Events (366,890), Users (2,779), CRM Activities (3,477), CRM Companies (500)	6 months period (Jan-Jun 2024), 500 unique customers
Data Processing	Missing value imputation, Date standardization, Outlier handling, Duplicate removal	Median imputation for numeric, Most frequent for categorical
Feature Engineering	Usage intensity score, Active user ratio, Engagement gap, Renewal urgency, Risk score	All features normalized to 0-100 scale

Section	Details	Notes
Model Training	Random Forest (100 trees), Gradient Boosting (100 estimators), Logistic Regression	Stratified 80-20 train-test split, Class weighting for imbalance
Evaluation Metrics	Accuracy, Precision, Recall, F1 Score, ROC AUC, Confusion Matrix	Focus on F1 Score for balanced performance
Tools & Libraries	Python, Pandas, Scikit-learn, Matplotlib, Seaborn, SQLite	Streamlit for dashboard, ReportLab for PDF generation

🔧 Implementation Details

📊 Data Pipeline

- SQLite database integration
- Automated data quality checks
- Incremental data processing
- Audit trail for all transformations

🤖 Model Architecture

- Random Forest: 100 trees, max depth 10
- Gradient Boosting: 100 estimators
- Logistic Regression with L2 regularization
- 5-fold cross-validation

📈 Performance Metrics

- Best model: Random Forest (F1: 0.648)
- Precision: 80%, Recall: 55%
- ROC AUC: 0.923
- Business value: $1.4M potential savings